

SYAM SUNDAR AKINA | AI ENGINEER

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Summary

- AI/ML Engineer with 5 years of experience building and deploying production-grade machine learning and generative AI systems across multiple domains.
- Proven expertise in large language models, retrieval-augmented generation (RAG), and multi-agent systems, delivering measurable improvements in accuracy, reliability, and user experience.
- Skilled in fine-tuning and optimizing models (LoRA, QLoRA, PEFT), designing evaluation frameworks, and ensuring fairness and explainability with SHAP, LIME, and bias mitigation practices.
- Experienced in developing cloud-native AI pipelines on AWS, GCP, and Azure, with deployments managed through Docker, Kubernetes, CI/CD, and MLflow.
- Strong background in NLP, semantic search, and computer vision, integrating knowledge graphs and vector databases for structured reasoning and scalable retrieval.
- Collaborative and impact-driven, with a track record of launching AI solutions that achieved 30% higher accuracy, 35% faster resolutions, and 40% gains in workflow efficiency.

Skills

- **AI & Machine Learning:** Large Language Models (GPT-4, LLaMA, Mistral, Claude, Amazon Titan), Transformer Architectures, NLP Pipelines, Semantic Search, Computer Vision, Recommendation Systems, Time-Series Forecasting, Anomaly Detection, Reinforcement Learning, Transfer Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation & Optimization
- **Generative AI & Advanced Techniques:** Retrieval-Augmented Generation (RAG), LoRA/QLoRA/PEFT Fine-Tuning, Prompt Engineering, Multi-Agent Systems (LangChain, LangGraph, CrewAI), Context-Aware Reasoning, Knowledge Graphs (Neo4j), Vector Search (FAISS, Pinecone, ChromaDB), Synthetic Data Generation
- **Responsible AI & Evaluation:** Bias Mitigation, Fairness Testing, Model Explainability (SHAP, LIME), Drift Detection, Prompt Injection Defense
- **MLOps & Deployment:** Docker, Kubernetes, MLflow, DVC, Airflow, Prefect, FastAPI, Flask, REST API Development, Terraform, Serverless (AWS Lambda), CI/CD for ML, Model Monitoring, Experiment Tracking
- **Programming & Scripting:** Python, SQL, Bash, TypeScript, R, YAML, JSON
- **Data Engineering & Storage:** PostgreSQL, MongoDB, Redis, Amazon S3, Snowflake, BigQuery, MySQL, SQLite, NoSQL, GraphQL, Knowledge Graph Databases
- **Analytics & Visualization:** Pandas, NumPy, Matplotlib, Tableau, Power BI, Exploratory Data Analysis (EDA), A/B Testing
- **Cloud Platforms:** AWS, GCP, Azure – Cloud-Native AI Pipelines

Experience

AI/ML Engineer | DeepMind

Jan 2025 – Present

California, USA

- Designed and deployed retrieval-augmented generation pipelines using large-scale vector search and approximate nearest neighbor (ANN) retrieval, improving accuracy of generative outputs by 30% and reducing response latency.
- Built knowledge graph-driven reasoning systems that increased onboarding task completion rates by 18% while providing transparent and explainable recommendations.
- Developed multi-agent systems for multimodal workflows such as template selection and dynamic page assembly, cutting setup time by 40% and improving user adoption.
- Led the end-to-end ML lifecycle including data preprocessing, fine-tuning with LoRA/QLoRA, hyperparameter optimization, and distributed deployment, maintaining 99% uptime across multiple AI modules.
- Created evaluation frameworks and custom benchmarks to track factual accuracy, bias, and latency, ensuring models consistently met production standards.
- Collaborated with product and design teams to integrate AI features that launched successfully in two production applications.

Data & AI Engineer Intern | IBM

Oct 2024 – Dec 2024

New York, USA

- Designed and implemented data pipelines for structured and unstructured datasets, enabling reproducible ML workflows and high-quality training data.
- Built time-series forecasting models (Prophet, ARIMA) that reduced prediction errors by 20% and improved operational efficiency.
- Developed NLP pipelines for text classification and sentiment analysis using Hugging Face and spaCy, improving classification accuracy on business-critical datasets.
- Designed and labeled evaluation datasets to benchmark NLP models, ensuring robustness and fairness before deployment.
- Created interactive dashboards in Tableau and Power BI to monitor KPIs and model performance in real time.
- Collaborated across engineering and analytics teams to integrate AI workflows into IBM Cloud Pak, streamlining production deployment

Programmer Analyst | Cognizant Technology Solutions

Dec 2021 – Jun 2023

Hyderabad, India

- Designed and executed 200+ dynamic marketing campaigns with real-time REST API triggers, impacting multi-million user datasets.
- Built Python-based segmentation models (K-means, Decision Trees) to target high-engagement customers, improving CTRs.
- Automated workflows with SQL and JavaScript, cutting campaign execution time by 40%.
- Developed churn prediction models integrated into real-time targeting engines, improving retention rates.
- Built Power BI dashboards to track ROI and AI-driven personalization metrics.
- Optimized data preprocessing pipelines for large-scale marketing datasets, reducing model training time by 35% and enabling faster A/B testing cycles

AI/ML Intern | DataVision Labs

Mar 2020 – Oct 2021

Pune, India

- Built predictive analytics models for user engagement forecasting, improving lead conversion rates by 25%.
- Automated e-commerce SKU metadata updates in Python, reducing manual work by 70%.
- Implemented NLP-based keyword clustering using TF-IDF to enhance ad targeting strategies.

- Developed churn prediction models in Scikit-learn to identify at-risk customers for proactive outreach.
- Conducted behavioral analytics using Google Analytics and Data Studio to improve marketing funnel efficiency.

Education

Master of Science in Information Systems and Technologies University of North Texas, Denton, TX	May 2025
Bachelor of Technology in Mechanical Engineering Bapatla Engineering College, India	Jul 2020

Projects

California Housing Price Prediction – End-to-End ML Workflow

- Python, scikit-learn, pandas, Streamlit, Jupyter
- Designed and deployed a full ML pipeline for predicting California housing prices using structured census data.
 - Performed data cleaning, EDA, and feature engineering to identify socioeconomic and geographic drivers.
 - Trained and optimized Linear Regression and Random Forest models with cross-validation, improving RMSE by 15%.
 - Delivered a Streamlit-based interactive app for real-time predictions, ready for SQL/Cloud API integration.

ResolveAI – Autonomous IT Workflow Agent

- LangGraph, FastAPI, FAISS, OpenAI
- Developed a multi-agent AI system (Coordinator, Diagnostic, Writer agents) for automating IT workflows at scale.
 - Integrated RAG pipeline with FAISS for real-time contextual retrieval and persistent memory.
 - Implemented approval workflows, retry logic, and fallback reasoning for production-grade reliability.
 - Delivered robust, explainable automation reducing IT issue resolution times by 35%.

Certifications

- Azure AI Engineer Associate Microsoft	2025
- DeepLearning.AI – Machine Learning Specialization Coursera	2025